

<div class = container >

<h1> Hello </h1>

</div>

JavaScript : → It is the programming language of the web. ^{10-15 LPA}

* It add interactivity ("clicking a button", "validating a form") JS "perform some animation"

⇒ (MERN Stack) ^{Express JS → Node JS} JS ^{CRUD} ^{4 names} ^{60-15 grp} (Useful)

Java → 20yrs ^{10-20 LPA}
+ HTML + CSS + JS
+ SQL → Backend
+ Server Side Prgmng
+ Spring Boot

<html>
 <head>
 <style> CSS </style>
 </head>
 <body>
 Everything here get display on the browser
 JavaScript code </script>
 </body>
 </html>

DOM → Document Object Model (EC / EEE) (FE) → DOM Structure

JavaScript can be written inside a <script> tag or a separate js file or in the Console Tab of the Browser directly.

1. script tag
2. separate script.js file
3. Console Tab in Browser [F12]

Important Concepts in JavaScript : →

1. Promise : →

- It means I will give result in future.
- It represents a task not finished yet.
- It has 3 states → pending, resolved, rejected
- Example → "Pizza is being prepared"

2. .then() →

- .then() means "After result do this"
- It runs after Promise is completed.
- Example → "When Pizza arrives eat it"

3. async →

- async makes a function → work with Promises easily.
- It assures that it will always return a Promise.
- Think of it as → "this function can wait"

4. await →

- await means → "Wait here until result comes"
- It pauses only that function, not the whole program.
- It only works inside "async" functions.

Most Imp Line : → (for interviews)

* "await only blocks the function, not the application".

Summary : →

Simple Analogy : →

Promise → Future result

then() → What to do later

await → Wait here

Promise → Order Placed

then() → Eat when Delivered

await → {
 Either sit & wait or
 watch TV or listen to
 music, etc.

The external js file like app.js or script.js is always added like this:

<html>
 <body>

Always here ⇒ <script src = "script.js"> </script>
 </body>
</html>

ChatGPT ⇒ Notes, Logic, etc, concepts

- Copilot ⇒ Code (Prompt)

Deepseek ⇒ Code + Large Content

(Paid) Claude.ai ⇒ Forget Coding

↓
opus

Cursor ⇒ Coding

Gemini ⇒ Coding/Concepts Nano Banana

AI Images (Midjourney, Grok)

MySQL Server

MySQL WorkBench 8.0.31
or above

MySQL Command Line Client

remember → username: root

→ password: 1234

D6 → JDBC
X

(Expense Tracker Project)

→ 11:30 SQL 4 hrs

→ D4 → SQL 6 hrs

10 hrs

+

2 hrs